

CDD-Net: Consistency-Discrepancy Decoupling for Chinese Pragmatic Intention Recognition

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Abstract

Existing multimodal spoken language understanding methods primarily emphasize cross-modal consistency fusion. Although effective for congruent text-speech interactions, this paradigm is often insufficient for deep pragmatic phenomena such as rhetorical questions, sarcasm, and indirect expressions. In such cases, speaker intention depends not only on literal content but also on fine-grained patterns of consistency and conflict between lexical semantics and vocal realization. A single fused representation therefore tends to obscure precisely the cues that matter most. To address this limitation, we propose the Consistency-Discrepancy Decoupling Network (CDD-Net), which explicitly models both cross-modal consistency and discrepancy. CDD-Net comprises: 1) Token-Level Discrepancy Localization (TLDL), which identifies conflict regions at the token-frame level; 2) Dual-Space Contrastive Learning (DSCL), which organizes evidence into separate consistency and discrepancy spaces; 3) Discrepancy-Guided Contrastive Pairing (DGCP) and Consistency-Discrepancy Reconstruction (CDR), which improve the discriminability and completeness of discrepancy-aware representations; and 4) a Raw Fusion Residual (RFR) path, which stabilizes optimization and improves robustness. We further introduce Tone-Aware Conflict Masking (TACM), a lightweight tone-aware refinement inside TLDL tailored to Chinese speech, and construct a new Chinese speech-text dataset together with a dedicated benchmark spanning both shallow/deep intention discrimination and fine-grained speech act recognition. Under a unified evaluation protocol, CDD-Net achieves the strongest SR performance among the reproduced baselines considered in this work, while additional PQP results suggest that the contribution of auxiliary cues such as explicit F0 is task-dependent. Overall, the findings indicate that explicitly modeling cross-modal discrepancy, rather than only cross-modal consistency, is beneficial for Chinese pragmatic spoken language understanding. Our code¹ and dataset² are publically available in the repository.

CCS Concepts

• **Computing methodologies** → **Natural language processing**: *Speech recognition; Machine learning approaches.*

Keywords

Pragmatic Spoken Language Understanding, Chinese Pragmatic Intention Recognition, Multimodal Learning, Cross-Modal Conflict Modeling, Consistency-Discrepancy Decoupling, Paralinguistic Cues

1 Introduction

Spoken language understanding (SLU) relies on the interplay between textual semantics and acoustic realization [9, 11, 12]. This

issue is particularly salient in Mandarin Chinese, where prosody can substantially alter the intent of otherwise identical utterances [13]. Pragmatic Intent Recognition (PIR) therefore requires jointly decoding semantic content and non-verbal realization to handle rhetorical questions, sarcasm, and indirect expressions [13, 26].

Existing multimodal SLU methods [3, 6, 14, 15] mainly model cross-modal consistency. This is insufficient for pragmatic cases where cues are localized, functionally heterogeneous, and often carried by conflict rather than agreement [16, 17, 19]. Existing disentanglement methods [4, 18] also focus on modality identity rather than pragmatic consistency versus discrepancy.

Figure 1 provides an intuition-building comparison between conventional multimodal fusion and our discrepancy-aware formulation. In pragmatic speech understanding, decisive evidence often comes from localized semantic-prosodic mismatches; conventional fusion tends to absorb these signals into a single shared representation, whereas our design preserves and sharpens conflict-bearing regions before prediction.

To address these limitations, we frame PIR from the perspective of consistency-discrepancy modeling and propose the Consistency-Discrepancy Decoupling Network (CDD-Net). This conceptual framework is realized through three primary architectural components: 1) Token-Level Discrepancy Localization (TLDL), which identifies cross-modal conflicts [3, 37]; 2) Dual-Space Contrastive Learning (DSCL), which organizes evidence into distinct consistency and discrepancy spaces [20]; and 3) a Raw Fusion Residual (RFR) path for more robust decision-making [22]. In addition, we incorporate Discrepancy-Guided Contrastive Pairing (DGCP), Consistency-Discrepancy Reconstruction (CDR), and an optional Tone-Aware Conflict Masking (TACM) mechanism as optimization and localization refinements within the training objective and TLDL pipeline [21].

To facilitate research, we also construct a new Chinese speech-text dataset and establish a benchmark covering both coarse-grained intent discrimination and fine-grained speech act recognition. Under a unified evaluation protocol, CDD-Net delivers the strongest overall results on the primary SR task among reproduced baselines such as MulT [3], MISA [4], and CCMT [5], while supplementary PQP results indicate that the contribution of auxiliary cues can remain task-dependent.

In summary, the primary contributions of this work are as follows:

- We frame Chinese pragmatic intent recognition as a matter of consistency-discrepancy modeling.
- We propose CDD-Net, combining TLDL, DSCL, DGCP/CDR, and an RFR path for robust discrepancy-aware reasoning.
- We construct a new Chinese speech-text dataset and establish a benchmark for pragmatic SLU, showing the benefits of explicit discrepancy modeling.

¹Code: <https://anonymous.4open.science/r/code111/README.md>

²Demo: <https://anonymous.4open.science/w/demo-F3C9/>

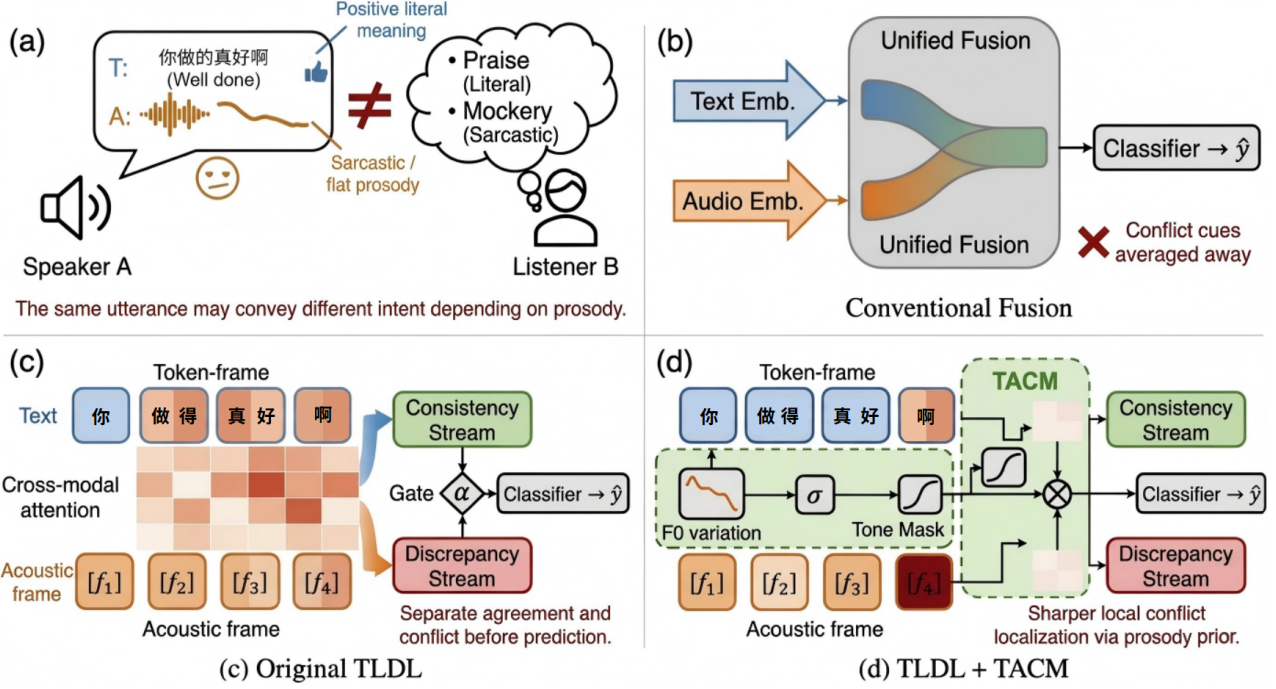


Figure 1: Comparison between conventional multimodal fusion and the proposed discrepancy-aware modeling strategy. In pragmatic intention recognition, crucial evidence often lies in localized semantic-prosodic mismatches. Conventional fusion tends to compress agreement and conflict into a single shared representation and can therefore weaken these cues; in contrast, TLDL explicitly localizes cross-modal conflict and TACM further sharpens conflict concentration through prosody-aware guidance.

- We introduce TACM, a lightweight tone-aware enhancement inside TLDL.

2 Related Work

2.1 Pragmatic and Spoken Language Understanding

Pragmatic understanding depends on both lexical semantics and paralinguistic expression [23, 24]. While speech cues are known to be crucial for intent recovery [25], widely used benchmarks such as IEMOCAP, MOSI, and MOSEI focus on English affective settings rather than Chinese speech-text pragmatics [8–10]. Chinese resources jointly covering literal-versus-deep interpretation and fine-grained speech acts remain scarce [13, 26].

2.2 Multimodal Fusion Models

Existing multimodal models mainly fall into three paradigms:

- **Fusion via Feature Interaction:** Utilizes feature concatenation and gating mechanisms to combine modalities. While effective, they typically assume that all modalities can be fused into a shared decision space without identifying whether the interaction reflects consistency or contradiction [6, 14, 27].
- **Attention-based and Transformer-based Fusion:** Employs cross-modal attention. MulT [3] and CCMT [5] model

temporal interactions explicitly [28] but still treat multimodal fusion as a fundamentally monolithic representation process [29].

- **Representation Disentanglement:** Utilizes representation decomposition to separate shared and modality-specific factors, as seen in MISA [4] and MAG-BERT [6, 18].

These methods still focus on modality identity more than pragmatic agreement versus conflict, although the latter is often the key signal for intent recognition [19].

2.3 Consistency, Incongruity, and Conflict Modeling

Recent work has also explored consistency and incongruity [30], but mostly in image-text alignment [38] or sentence-level contradiction [31]. Speech-text pragmatics differs because conflicts are often local and structurally meaningful [13], motivating TLDL and the dual-space design of CDD-Net.

3 Task Definition and Benchmark

3.1 Motivation for a New Benchmark

While pragmatic understanding heavily relies on the interplay between lexical semantics and paralinguistic expressions [23, 24], most established multimodal benchmarks (e.g., IEMOCAP [10],

MOSI [8]) are centered on English affective tasks. In contrast, comparable Chinese resources that jointly cover literal-versus-deep interpretation and fine-grained speech act recognition remain limited [13, 26]. To address this gap and specifically investigate pragmatic consistency and discrepancy in Chinese, we construct a dedicated two-task benchmark.

3.2 Benchmark Construction and Tasks

Our benchmark is built around paired Chinese speech-text inputs. We design 84 polarity questions, each paired with a neutral context (ISQ) and a negative/rhetorical context (RQ), yielding 168 context-question prompts across four everyday scenarios: marriage, workplace, school, and family. Data collection follows a controlled-yet-natural pipeline with scenario design, question-response elicitation, smartphone-based self-recording, and transcription normalization. We retain only valid Mandarin recordings with intact question-response pairing and usable annotation files. Intent labels follow a shared guideline centered on the responder's primary focus, including factual/informational content, attitude, emotion, commitment, and discourse extension. To reduce obvious label inconsistency, ambiguous cases were manually revisited and checked before finalizing the annotations.

The benchmark contains two tasks:

- **PQP (Pragmatic Question Pair):** A binary boundary-analysis task distinguishing literal versus deep pragmatic meaning (e.g., whether an utterance is a straightforward question or carries rhetorical intent).
- **SR (Speech Act Recognition):** A 14-class fine-grained intention recognition task, grouped into five super-categories (Table 1).

PQP captures boundary conditions for speech necessity, while SR provides the main test bed for structured discrepancy modeling.

Table 1: SR intent taxonomy.

Category	Labels	Short description
Literal/Factual	F, A, T, W	factual confirmation, contradiction, conditional answer, or knowledge/suggestion-based response
Attitude	S, D, R	submission/apology, disagreement, or response to dissatisfaction involving a third party
Emotion	E, V	emotional support or perfunctory response
Commitment	I, H	promise or expression of anticipation/desire
Extension	C, K, M	topic shift, continuation of the original topic, or counter-question

3.3 Data Split and Features

Both PQP and SR use strict speaker-level train/dev/test splits to prevent speaker memorization. The benchmark contains recordings from 209 native Mandarin speakers in total, including 103 speakers in PQP and 106 speakers in SR. Table~2 summarizes the sample statistics for both tasks.

Table 2: Dataset statistics.

Item	PQP	SR
Total samples	4,116	6,580
Train split	~3,233	~5,214
Dev split	~389	~659
Test split	~494	~707
Number of labels	2	14

For SR, each response can be paired with its eliciting *question* from PQP, enabling optional context-grounded conditioning. We intentionally construct a heterogeneous feature set to comprehensively capture the diverse signals that jointly determine pragmatic intention in spoken Chinese. From a linguistic perspective, textual semantics provide the structural and literal foundation of the utterance. From an acoustic perspective, global spectral and energy features characterize the overall speaking style and macroscopic emotional state. Crucially, from a prosodic perspective, local variations—such as pitch inflections, intensity shifts, and rhythm alterations—encode fine-grained pragmatic markers (e.g., irony, rhetorical emphasis, or hesitation). Because true pragmatic intent often arises from the friction between literal semantics and these localized prosodic events, relying on a unimodal source frequently results in missing key cross-modal conflicts [13, 23, 24]. Therefore, each sample encompasses:

- **Text:** captures propositional content and lexical meaning, which provides the semantic anchor for pragmatic interpretation [1, 24].
- **Audio backbone features:** capture utterance-level speaking style and latent prosodic patterns from self-supervised acoustic representations, complementing text when literal meaning is ambiguous [2, 23].
- **Explicit F0:** provides continuous pitch statistics that directly relate to intonation and rhetorical markedness in Chinese speech [13, 32].
- **SpeechCraft labels:** provide coarse symbolic categories for pitch, energy, and speed, improving robustness when raw contours are noisy or speaker-dependent [33].
- **Frame-level acoustics:** provide local low-level cues that can align with token-level pragmatic conflict regions, supporting fine-grained discrepancy localization [34, 37].

This design is not a simple feature accumulation: each cue family is assigned a distinct role in the consistency-discrepancy pipeline (semantic anchoring, global style modeling, prosody-sensitive conflict emphasis, and local conflict localization), and later ablations (Sec. 5.4) verify that removing specific components leads to predictable performance drops. The benchmark is challenging because text ambiguity is high on PQP, SR is imbalanced, and many decisions depend on semantic-vocal conflict. The corpus provides waveform, TextGrid, and PitchTier annotations at 44.1 kHz, together with transcriptions and intent labels. In experiments, PQP serves as a boundary-analysis task, while SR is the primary evaluation task.

4 Method

4.1 Overview

CDD-Net treats multimodal pragmatic understanding as a structured consistency-discrepancy modeling problem. Let an input utterance be represented as (x', x^a) , where x' is the text sequence and x^a is the speech signal, together with optional auxiliary cues. The target is to predict a pragmatic label y by jointly reasoning about semantic content, vocal realization, and their local interaction pattern.

The model proceeds in four stages:

- (1) **Encoding:** textual and acoustic inputs are encoded into sequence-level and utterance-level representations.
- (2) **Discrepancy localization:** TLDL estimates where text and speech conflict at token-frame granularity.
- (3) **Dual-space organization:** DSCL, DGCP, and CDR encourage the two spaces to specialize without collapsing.
- (4) **Robust decision fusion:** RFR combines structured decoupling with a raw-fusion residual path.

Figure 2 shows the SR-oriented CDD-Net with RFR architecture.

4.2 Input Encoding

We employ Chinese BERT [1] and Wav2Vec2 [2] to extract contextualized text ($T_{\text{seq}} \in \mathbb{R}^{L_t \times d}$) and acoustic ($A_{\text{seq}} \in \mathbb{R}^{L_a \times d}$) representations. We also use explicit F0 [32], symbolic SpeechCraft labels [33], and low-level frame acoustics [34], all projected into a unified latent space [35]. For PQP, we keep the same backbone but remove SR-specific DGCP/CDR/RFR components.

4.3 Token-Level Discrepancy Localization

TLDL pinpoints *where* cross-modal conflict occurs. Since pragmatic intent often hinges on local semantic-vocal mismatches rather than global tone [36], it uses cross-modal attention [37] to generate fine-grained discrepancy scores over text tokens and acoustic frames.

Let $T_{\text{seq}} \in \mathbb{R}^{L_t \times d}$ and $A_{\text{seq}} \in \mathbb{R}^{L_a \times d}$ denote the sequence representations for text and audio, respectively. To pinpoint fine-grained mismatches, TLDL first captures token-frame interactions via bidirectional cross-modal attention, producing cross-attended features. These interaction signals are subsequently transformed into localized discrepancy scores S_t :

$$\begin{aligned} S_t^T &= \sigma \left(\frac{f_{\text{text}}(\text{CrossAttn}(T_{\text{seq}}, A_{\text{seq}}))}{\tau} \right), \\ S_t^A &= \sigma \left(\frac{f_{\text{audio}}(\text{CrossAttn}(A_{\text{seq}}, T_{\text{seq}}))}{\tau} \right) \end{aligned} \quad (1)$$

where f_{text} and f_{audio} are projection networks, σ is the sigmoid activation, and τ is a learnable temperature parameter to control the sharpness of the discrepancy distribution. This mechanism avoids relying solely on holistic utterance-level judgments by assigning an explicit conflict weight to each token and acoustic frame.

The discrepancy scores act as soft gates that partition the original representations into discrepancy-oriented and consistency-oriented components. For instance, the text discrepancy representation is

$$T_{\text{disc}} = \frac{1}{\sum_i S_{t,i}^T m_i^T + \epsilon} \sum_i (S_{t,i}^T m_i^T) T_{\text{seq},i} \quad (2)$$

where m_i^T denotes the valid sequence mask. An analogous operation yields A_{disc} , T_{cons} , and A_{cons} .

To capture global semantic shift, we compute macro-level discrepancy vectors (T_{disvec} , A_{disvec}) as residual differences between the original modality summaries and their cross-modally attended counterparts:

$$\begin{cases} T_{\text{disvec}} = \text{Pool}(T_{\text{seq}}) - \text{Pool}(\text{CrossAttn}(T_{\text{seq}}, A_{\text{seq}})), \\ A_{\text{disvec}} = \text{Pool}(A_{\text{seq}}) - \text{Pool}(\text{CrossAttn}(A_{\text{seq}}, T_{\text{seq}})). \end{cases} \quad (3)$$

Here, $\text{Pool}(\cdot)$ denotes the utterance-level summarization operator. These vectors guide the discrepancy stream and the adaptive combiner.

TACM. To better adapt TLDL to Chinese pragmatic speech understanding, we introduce Tone-Aware Conflict Masking (TACM), which reweights discrepancy scores using a PitchTier-derived F0-variation proxy. Let ΔF_0 denote the normalized F0-variation signal extracted from PitchTier annotations. We first compute a tone-aware mask

$$M_{\text{tone}} = \sigma(\Delta F_0 / T_{\text{tone}}), \quad (4)$$

where T_{tone} is a temperature hyperparameter. The original discrepancy scores are then modulated as

$$\begin{aligned} \tilde{S}_t^T &= \text{clip}(S_t^T \odot (1 + \gamma_{\text{tone}} M_{\text{tone}}), 0, 1), \\ \tilde{S}_t^A &= \text{clip}(S_t^A \odot (1 + \gamma_{\text{tone}} M_{\text{tone}}), 0, 1), \end{aligned} \quad (5)$$

where γ_{tone} controls the strength of tone-aware reweighting and the mask is broadcast to the token and frame dimensions. In the current implementation, we use utterance-level F0 standard deviation as the default proxy.

Figure 3 illustrates the TACM pipeline. This design encourages the model to preserve localized inconsistency as a decision-relevant signal rather than averaging it away in a monolithic multimodal representation.

4.4 Dual-Space Contrastive Learning

Local discrepancy localization alone is insufficient because the resulting representations may still remain entangled. DSCL explicitly organizes two functional spaces:

- a **consistency space**, which captures evidence jointly supported by text and speech; and
- a **discrepancy space**, which captures mismatch patterns, pragmatic markedness, and intention-discriminative conflict cues.

$$\begin{cases} c_{\text{out}} = f_{\text{cons}}([T_{\text{cons}} \parallel A_{\text{cons}}]), \\ d_{\text{repr}} = f_{\text{disc}}([T_{\text{disc}} \parallel A_{\text{disc}} \parallel \text{prosody}_{\text{emb}} \parallel \text{sc}_{\text{emb}} \parallel \text{fa}_{\text{emb}}]). \end{cases} \quad (6)$$

Here, f_{cons} and f_{disc} denote learnable projection functions for the two semantic roles.

We fuse these specialized representations with a dynamic gate guided by the macro-level discrepancy vectors:

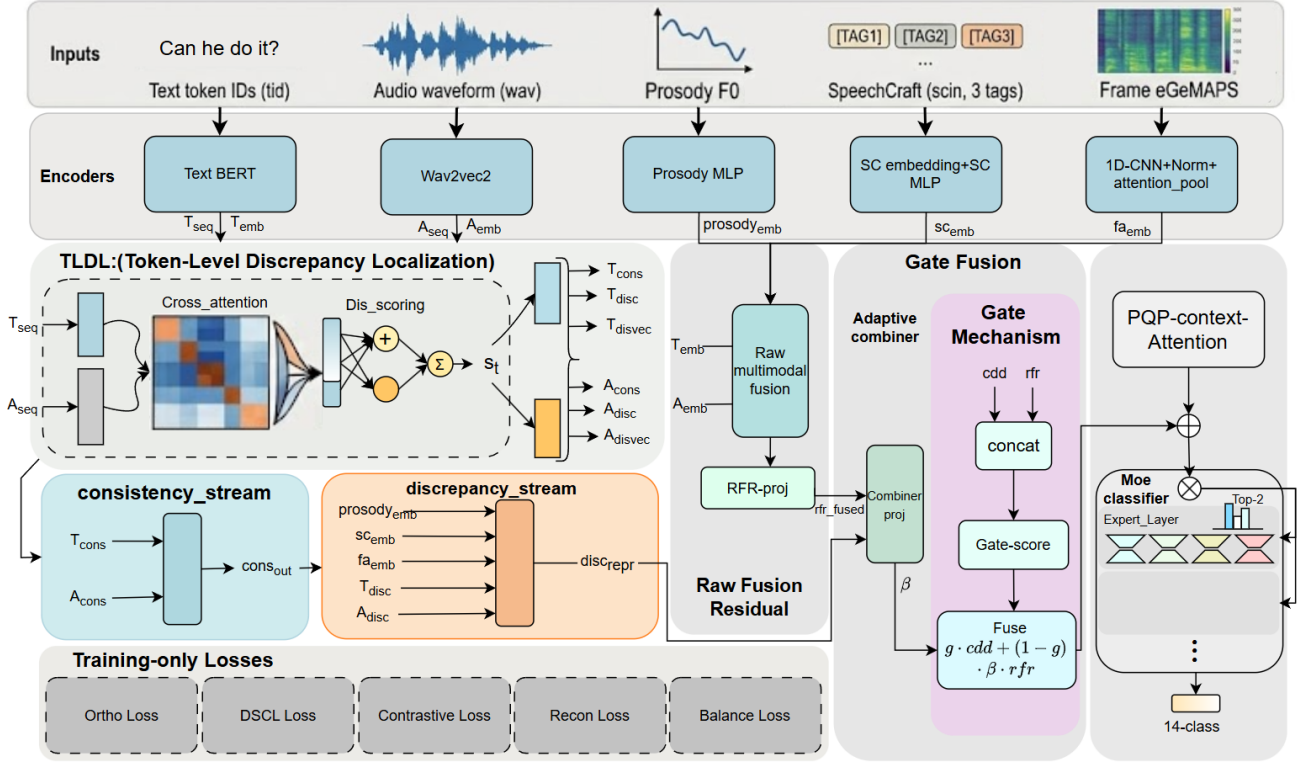


Figure 2: SR-oriented CDD-Net. The model encodes multimodal inputs, localizes token-frame discrepancy, organizes consistency/discrepancy streams, and combines them with a raw-fusion residual path.

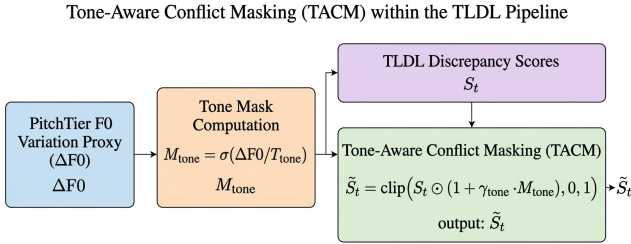


Figure 3: TACM. PitchTier-derived F0 variation is converted into a tone-aware mask for reweighting TLDL discrepancy scores.

$$\begin{aligned} \mathbf{h}_{disvec} &= [\mathbf{T}_{disvec} \parallel \mathbf{A}_{disvec}], \\ \alpha &= \sigma(W_\alpha \mathbf{h}_{disvec} + b_\alpha), \\ \mathbf{h}_{cdd} &= \alpha \mathbf{c}_{out} + (1 - \alpha) \mathbf{d}_{repr}, \end{aligned} \quad (7)$$

where W_α and b_α parameterize the adaptive combiner.

To encourage specialization, the discrepancy representation is trained with a supervised contrastive objective, optionally reweighted

by discrepancy-pattern similarity:

$$\mathcal{L}_{con} = -\frac{1}{B} \sum_{i=1}^B \frac{1}{\sum_{p \neq i} w_{ip}} \sum_{p \neq i} w_{ip} \log \frac{\exp(\mathbf{d}_i^\top \mathbf{d}_p / \tau_s)}{\sum_{a \neq i} \exp(\mathbf{d}_i^\top \mathbf{d}_a / \tau_s)}, \quad (8)$$

where B is the batch size and $\mathbf{d}_i$ denotes the normalized discrepancy representation of sample i . When DGCP is disabled, $w_{ip} = 1$ for samples with the same class label and $w_{ip} = 0$ otherwise, yielding a standard supervised contrastive loss. When DGCP is enabled, the positive weight is further modulated by the similarity between discrepancy-score patterns:

$$w_{ip} = 1[y_i = y_p] \cdot ((1 - \gamma) + \gamma \cdot \text{sim}(\mathbf{s}_i, \mathbf{s}_p)), \quad (9)$$

where $\mathbf{s}_i$ is the token-level discrepancy-score vector, $\text{sim}(\cdot, \cdot)$ is cosine similarity rescaled to $[0, 1]$, and $\gamma \in [0, 1]$ controls the strength of discrepancy guidance. Equation 9 biases the positive set toward same-label pairs that also share similar local conflict patterns. DGCP is applied only to same-label pairs.

To prevent representational collapse, we incorporate auxiliary losses. The orthogonality term minimizes overlap between the consistency and discrepancy components:

$$\mathcal{L}_{ortho} = \mathbb{E}[(\mathbf{t}_{cons}^\top \mathbf{t}_{disc})^2] + \mathbb{E}[(\mathbf{a}_{cons}^\top \mathbf{a}_{disc})^2]. \quad (10)$$

DSCL further uses two regularizers:

$$\mathcal{L}_{align} = 1 - \cos(\mathbf{t}_{cons}, \mathbf{a}_{cons}), \quad \mathcal{L}_{sep} = \cos(\mathbf{c}_{out}, \mathbf{d}_{repr})^2, \quad (11)$$

They encourage cross-modal alignment in the consistency space and separation between final consistency and discrepancy representations.

The overall objective combines the primary classification loss with these auxiliary terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}} + \lambda_{\text{orth}} \mathcal{L}_{\text{ortho}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{sep}} \mathcal{L}_{\text{sep}} + \lambda_{\text{recon}} \mathcal{L}_{\text{Recon}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}}, \quad (12)$$

where $\mathcal{L}_{\text{bal}}$ is a small expert-balancing term used only when the MoE classifier is enabled. Equation 12 summarizes the full optimization target. In the implementation, it penalizes highly skewed expert usage and is added with a fixed coefficient λ_{bal} together with the primary classification loss (so $\mathcal{L}_{\text{CE}}$ in practice denotes that summed supervised objective when MoE is enabled). In training, the weighted CDD terms $\lambda_{\text{con}} \mathcal{L}_{\text{con}}$ through $\lambda_{\text{recon}} \mathcal{L}_{\text{Recon}}$ are linearly warmed up over the first few epochs to stabilize optimization (Sec. 5.2).

This split is functional rather than modal: a token or frame may contribute to either space depending on whether it reinforces or contradicts the pragmatic interpretation. Overall, DSCL learns a structured latent space in which agreement and conflict play different semantic roles.

4.5 Discrepancy-Guided Contrastive Pairs and Reconstruction

We further add two collaborative components.

DGCP reweights positive pairs according to discrepancy-pattern similarity, since two utterances may share a label while realizing it through different local conflict configurations.

CDR constrains the two streams to preserve complementary information from the original multimodal input. The reconstruction loss is

$$\mathcal{L}_{\text{Recon}} = \mathbb{E} \left[\left\| \text{MLP}_{\text{recon}} ([\mathbf{c}_{\text{out}} \parallel \mathbf{d}_{\text{repr}}]) - \mathbf{h}_{\text{orig}} \right\|_2^2 \right] \quad (13)$$

where $\mathbf{h}_{\text{orig}}$ is the raw fused multimodal target. This term discourages the decoupling from discarding task-relevant information.

4.6 Raw Fusion Residual

Structured consistency-discrepancy modeling is expressive but can be sensitive to optimization dynamics and dev-test mismatch. We therefore introduce the Raw Fusion Residual (RFR), a direct multimodal fallback path.

The raw branch fuses text, audio, and auxiliary paralinguistic embeddings using a direct projection. The decoupled branch produces a structured representation from the consistency and discrepancy streams. A learnable gate then interpolates between the two:

$$g = \sigma \left(\frac{W_{\text{gate}} [\mathbf{h}_{\text{cdd}} \parallel \mathbf{h}_{\text{rfr}}]}{\tau_{\text{gate}}} \right), \quad (14)$$

$$\mathbf{h}_{\text{fused}} = g \cdot \mathbf{h}_{\text{cdd}} + (1 - g) \cdot \beta \cdot \mathbf{h}_{\text{rfr}},$$

where g is the learned gate, W_{gate} is a linear projection layer, and τ_{gate} is the gate temperature hyperparameter. β is a learnable scaling factor initialized to 1.0 and clamped between [0.1, 2.0] during training to prevent the raw path from overpowering the structured representation. $\mathbf{h}_{\text{cdd}}$ is the structured representation, and $\mathbf{h}_{\text{rfr}}$ is the raw residual representation.

RFR preserves a stable direct-fusion path when structured discrepancy modeling is unreliable, enabling a calibrated trade-off between expressiveness and robustness.

4.7 PQP-Grounded Context and Hierarchical MoE Head (SR)

The SR pipeline can incorporate paired PQP *question* evidence alongside the answer utterance. We encode the question side with the same BERT and Wav2Vec2 families, summarize it into a compact multimodal vector, and use a lightweight attention block to update the SR utterance representation. If PQP inputs are absent, this block is skipped.

The final 14-way classifier is a hierarchical mixture-of-experts head aligned with the five super-categories in Table 1 [42]. Each super-category defines one routing group with four experts and top-2 sparse routing, while a small load-balancing term discourages collapsed routing.

5 Experiments

5.1 Datasets and Baselines

We evaluate our framework on our newly constructed multimodal dataset, which comprises two tasks: Pragmatic Question Pair (PQP) for binary boundary analysis and Speech Act Recognition (SR) for 14-class fine-grained intention classification. Both tasks feature a challenging speaker-level train/dev/test split.

We compare CDD-Net against several strong baselines under a unified implementation setting:

- **Multimodal full:** a strong fusion baseline combining text, audio, prosody, SpeechCraft, and frame-level acoustics.
- **Multimodal -F0:** the same baseline without explicit prosody.
- **MuT (reproduced):** a cross-modal Transformer baseline [3].
- **MISA (reproduced):** a modality-invariant and modality-specific decomposition baseline [4].
- **CCMT (reproduced):** a cascaded cross-modal Transformer baseline [5] adapted to the same Chinese speech-text setting.

All reproduced baselines are trained under the same speaker-level split, backbone family, and closely matched optimization framework. We also align key non-architectural settings, including effective batch size, training horizon, shared feature stack, and checkpoint selection criteria, so that observed differences mainly reflect modeling choices.

5.2 Experimental Setup

We use BERT-base-Chinese as the text encoder and a Chinese Wav2Vec2 model as the acoustic encoder. On SR, we report accuracy, macro-F1, and weighted-F1; on PQP, we report accuracy and F1. The optional PQP-grounded context encoder and hierarchical MoE head described in Section 4.7 are not included in the standalone CDD variants reported in the main SR comparison and ablation tables unless explicitly noted. The training objective combines focal loss [7], R-Drop [40], mixup [39], and the CDD auxiliary losses introduced in Section 4. Detailed optimization, calibration, and loss-weight settings are summarized in Table 3. These values were chosen so that classification remains the dominant term while

Table 3: Key training hyperparameters.

Setting	Value
Encoders	BERT-base-Chinese / Chinese Wav2Vec2
Proj. dim	128
Batch size	32 ($4 \times$ accumulation 8)
Horizon	50 epochs, patience 15
Optimization	enc LR 1×10^{-5} , cls LR 5×10^{-4} , WD 1×10^{-2}
Stabilization	warmup 0.03, grad clip 1.0, label smooth 0.05
Enc. schedule	frozen 5 ep; partial audio unfreezing from ep 6
Tricks	Mixup $\alpha = 0.3$ (ep 6); R-Drop $\alpha = 1.0$ (ep 10); SWA (ep 28)
CDD weights	$\lambda_{\text{con}} = 0.01$, $\lambda_{\text{orth}} = 0.001$, $\lambda_{\text{align}} = 0.005$, $\lambda_{\text{sep}} = 0.001$, $\lambda_{\text{recon}} = 0.01$, $\lambda_{\text{bal}} = 0.01$
Aux settings	CDD warmup min(1, $e/5$); DGCP $\gamma = 0.5$
TACM	PitchTier F0 std; $\gamma_{\text{tone}} = 0.5$; $T_{\text{tone}} = 1.0$

Table 4: Comparison of different methods in SR tasks. Reproduced baselines use the same split and feature setting.

Method	Acc	M-F1	W-F1
Multimodal (full)	72.56	0.4337	0.6997
Multimodal (-F0)	70.01	0.4212	0.6683
MuT (reproduced)	70.30	0.4806	0.6840
MISA (reproduced)	72.84	0.4343	0.6946
CCMT (reproduced, SWA)	72.98	0.4965	0.7065
CDD+RFR	74.47 $\pm$ 0.82	0.4777 $\pm$ 0.0092	0.7191 $\pm$ 0.0103
CDD+RFR+TACM (SWA)	76.24$\pm$0.53	0.5560$\pm$0.0057	0.7417$\pm$0.0066

contrastive and reconstruction signals contribute at a moderate scale; we did not run an extensive grid search over them. We save checkpoints separately according to dev accuracy and dev macro-F1; for the calibrated SR setting, SWA [41] starts at epoch 28 and averages model states from epoch 28 onward. The main CDD variants in Table 4 and the SR ablation variants in Table 6 are each run with three independent seeds and reported as mean $\pm$ std, whereas reproduced baselines are shown with single scores when matched variance estimates are unavailable.

This setup emphasizes controlled comparison under a single internal protocol rather than cross-paper number aggregation.

5.3 Main Results

Table 4 reports the performance on the 14-class SR task. Multimodal methods consistently outperform unimodal baselines, confirming that lexical content alone is insufficient for resolving complex pragmatic intentions. Among existing architectures, CCMT serves as the strongest reproduced baseline.

The full CDD-Net (with RFR, TACM, and SWA) achieves state-of-the-art performance, reaching 76.24% accuracy and 0.5560 macro-F1. Given the inherent class imbalance in SR, we prioritize macro-F1 as it better reflects the model’s ability to capture the full intention space rather than overfitting to frequent patterns. These gains provide empirical evidence that explicit discrepancy modeling effectively captures fine-grained pragmatic cues.

Table 5 evaluates PQP as a supplementary boundary task. While CDD-Net remains competitive, the strongest results are observed

Table 5: Comparison of different methods in PQP tasks.

Method	Acc	F1
Text-only	0.5486	0.4297
Audio-only	0.6457	0.5843
Multimodal (full)	0.7794	0.7696
Multimodal (-F0)	0.8623	0.8528
CDD-Net	0.7591	0.7326
CDD-Net(-F0)	0.8360	0.8280

Table 6: SR ablation results (mean $\pm$ std over three independent runs).

Variant	Acc	M-F1	W-F1
Ours (Full)	76.24$\pm$0.53	0.5560$\pm$0.0057	0.7417$\pm$0.0066
w/o TACM	74.47 $\pm$ 0.82	0.4777 $\pm$ 0.0092	0.7191 $\pm$ 0.0103
w/o RFR	71.77 $\pm$ 0.73	0.4523 $\pm$ 0.0079	0.6976 $\pm$ 0.0053
w/o DSCL	72.17 $\pm$ 0.29	0.4747 $\pm$ 0.0046	0.7034 $\pm$ 0.0050
w/o TLDL	71.50 $\pm$ 0.43	0.4058 $\pm$ 0.0041	0.6973 $\pm$ 0.0057
w/o DGCP/CDR	70.87 $\pm$ 0.14	0.4205 $\pm$ 0.0044	0.6842 $\pm$ 0.0054

in the F0-ablated multimodal baseline (86.23% Acc). A plausible explanation is that PQP is driven more by utterance-level speaking style than by sharply localized token-frame conflicts. In this regime, the Wav2Vec2 backbone and the symbolic SpeechCraft prosody labels may already capture most of the decision-relevant prosodic information, so explicit continuous F0 provides limited additional signal. Moreover, raw F0 contours are more sensitive to speaker-specific variation and pitch-extraction noise, which can introduce feature redundancy or mild overfitting under a speaker-level split. We therefore interpret the PQP trend not as evidence that discrepancy modeling is ineffective in general, but rather as a task-dependent result: the full benefit of the framework is most clearly reflected on the fine-grained SR task, while PQP mainly serves as a supplementary motivation-oriented boundary test.

5.4 Ablation Study

To evaluate the contribution of individual components within CDD-Net, we conduct comprehensive module-removal ablations on the SR dataset. The quantitative results, averaged over three independent runs, are summarized in Table 6. Removing TLDL causes the most significant performance degradation, particularly in the macro-F1 score, confirming that local token-frame discrepancy localization is central to the architecture. The removal of DSCL and RFR also negatively impacts overall performance, while eliminating DGCP/CDR further weakens both macro-F1 and weighted-F1 metrics.

Qualitative Visualization. To intuitively understand the internal mechanisms of discrepancy modeling, we extract and visualize the attention weights and discrepancy scores generated by the TLDL module. Specifically, we compare the localization behavior with and without the TACM enhancement on representative audio-text segments. As illustrated in Figure 4, the base TLDL effectively concentrates discrepancy scores on pragmatically marked regions,

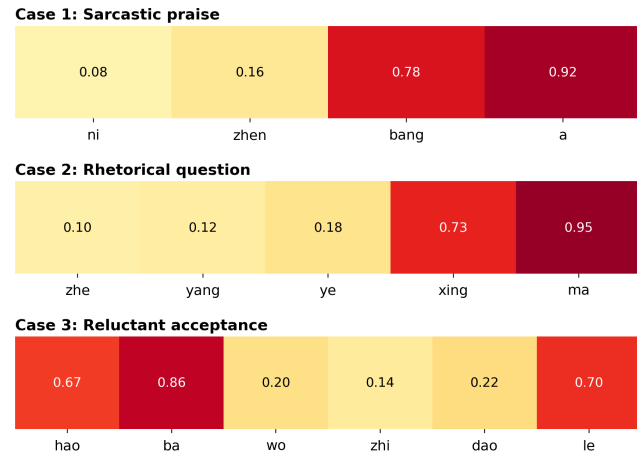


Figure 4: Illustrative TLDL heatmaps. Discrepancy scores concentrate on pragmatically marked regions.

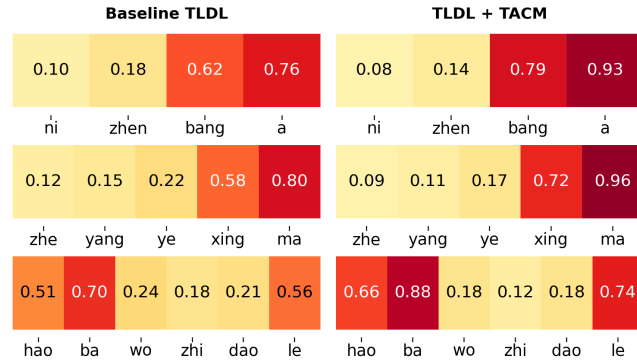


Figure 5: TLDL vs. TLDL+TACM. TACM yields sharper discrepancy concentration on prosodically marked regions.

capturing the primary cross-modal conflicts. Furthermore, the comparative visualization in Figure 5 demonstrates that incorporating TACM yields a sharper and more precise concentration of weights on prosodically salient segments. This confirms that tone-aware reweighting provides crucial language-specific priors, successfully guiding the model to focus on the most informative acoustic-lexical discrepancies.

Calibration Effects. We further investigate the impact of the Raw-Fusion Residual (RFR) calibration strategy by analyzing how gate temperature adjustments and Stochastic Weight Averaging (SWA) timing influence the routing between the structured branch and the raw fallback branch. The learning curves and calibration effects on the SR dataset are plotted in Figure 6. The analysis indicates that proper calibration significantly mitigates the discrepancy between validation and test performance (i.e., dev-test mismatch). By stabilizing the validation behavior during the later stages of training, the calibrated RFR mechanism prevents the model from overfitting to spurious correlations in the structured representation,

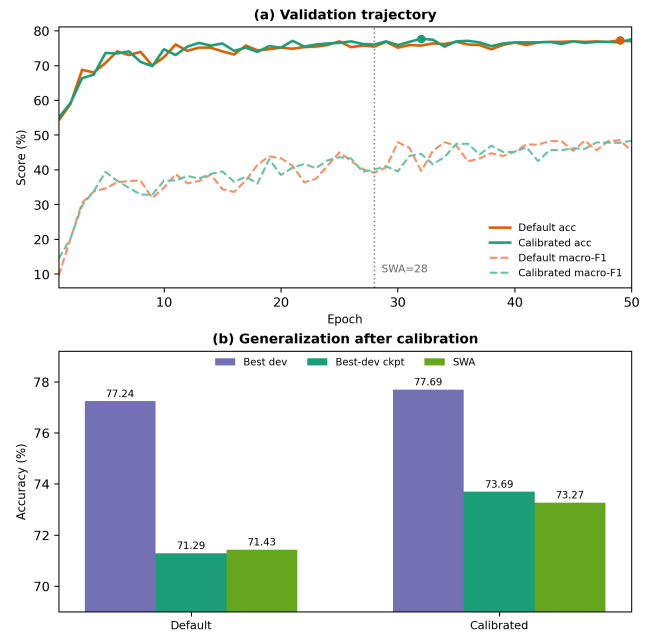


Figure 6: RFR calibration on SR. Calibrated training stabilizes validation behavior and improves final test accuracy after SWA.

ultimately leading to improved and more robust test-time accuracy after applying SWA.

6 Conclusion

We present CDD-Net for Chinese pragmatic intent recognition, framing the task as a consistency-discrepancy modeling problem to move beyond conventional, purely consistency-oriented fusion approaches. The framework integrates token-frame discrepancy localization, dual-space evidence organization, and a raw-fusion residual path to explicitly preserve conflict-aware cues while maintaining overall robustness. Furthermore, we introduce TACM as a lightweight, Chinese-oriented prosody refinement mechanism within TLDL, demonstrating that language-specific prosody priors can effectively sharpen local conflict localization. Beyond the architecture, we constructed a dedicated Chinese speech-text benchmark encompassing both shallow/deep intention discrimination and fine-grained speech-act recognition. While future investigations will extend this paradigm to accommodate more complex real-world conditions—such as ASR noise, speaker overlap, and broader conversational contexts—alongside deeper qualitative minority-class analyses, our current SR and PQP results strongly indicate that explicit discrepancy modeling constitutes a highly promising direction for pragmatic spoken language understanding.

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